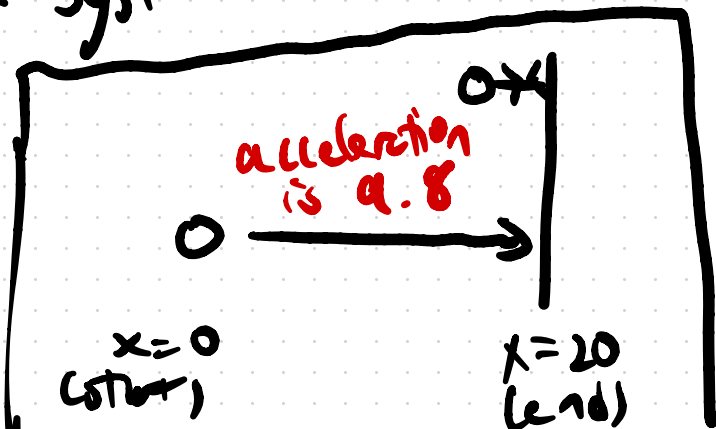
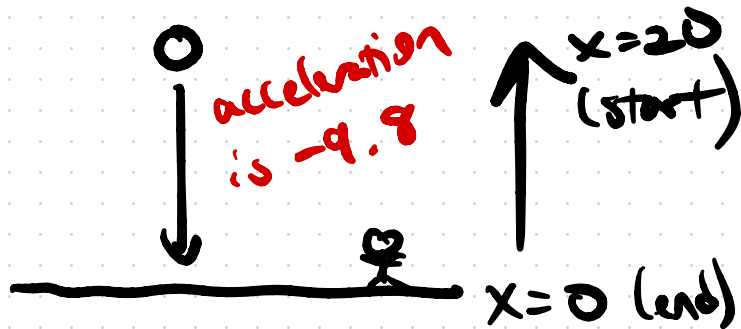


Ignoring air resistance, a falling object accelerates at 9.8 m/s^2 . An object is dropped from a height of 20 m . How fast is it moving when it hits the ground?

Strategy: Find expression for the velocity and position of the object, and figure the rest from there.

Need to decide our coordinate system: *we'll use this.*



$$a(t) = 9.8 \quad \text{acceleration} \quad (\text{use } -9.8 \text{ if using the other coordinate system})$$

$$v(t) = 9.8t + C \quad \text{velocity}$$

$$s(t) = 4.9t^2 + Ct + D \quad \text{position}$$

In the coordinate system we chose,

"object starts at rest" $v(0) = 0$

"object starts at $x=0$ " $s(0) = 0$

$$v(0) = 9.8 \cdot 0 + C$$

$$0 = C$$

$$s(0) = 4.9 \cdot 0 + C \cdot 0 + D$$

$$0 = D$$

If we chose the other coordinate system,

$$v(0) = 0$$

$$s(0) = 20$$

$$a(t) = 9.8 \quad v(t) = 9.8t \quad s(t) = 4.9t^2$$

We want the value of $v(t)$ when $s(t) = 20$.

Let T be time when $s(T) = 20$.

$$4.9T^2 = 20$$

$$T = \sqrt{\frac{20}{4.9}} \approx 2.02$$

$$v(T) = 9.8T \approx 19.80$$

What is the general form for an antiderivative of $f(x) = \frac{1}{x}$?

One antiderivative on $(0, \infty)$ is

$\ln x$ ← only accept positive x

General form on $(0, \infty)$

$$\ln x + C$$

$$\frac{d}{dx} \ln(-x) = \frac{1}{-x} = -\frac{1}{x}$$

One antiderivative on $(-\infty, 0)$ is $\ln(-x)$

General form on $(-\infty, 0)$ is $\ln(-x) + C$

Is $\ln|x| + C$ the general form for antiderivatives on $(-\infty, 0) \cup (0, \infty)$?

No. Consider

$$f(x) = \begin{cases} \ln x + C & \text{for } x > 0 \\ \ln(-x) + D & \text{for } x < 0 \end{cases}$$

where C, D are constants (possibly different)

Theorem: Let $f(x)$ be a function defined on an interval, and let $F(x)$ be any antiderivative of f on that interval. Then all antiderivatives of f on that interval are of the form.

$$G(x) = F(x) + C$$

where C is a real constant.

If we want an antiderivative defined on $(-\infty, 0) \cup (0, \infty)$, we need a different constant on each interval $(-\infty, 0)$, $(0, \infty)$.

On the interval $(0, \infty)$, the general form of an antiderivative of $\frac{1}{x}$ is

$$\ln x + C$$

On the interval $(-\infty, 0)$, the general form of an antiderivative of $\frac{1}{x}$ is

$$\ln(-x) + D$$

You will often see these combined as

$$\ln |x| + C$$

This should be understood as the form of all antiderivatives on $(0, \infty)$, or all antiderivatives on $(-\infty, 0)$, but not all antiderivatives defined on $(-\infty, 0) \cup (0, \infty)$ are of this form.

The Area Problem

(5.1)

Zeno's arrow flies through the air in a line. Its velocity at various times is given below.

Time (s)	0	0.1	0.2	0.3	0.4	0.5
Velocity (m/s)	100	80	70	65	60	60

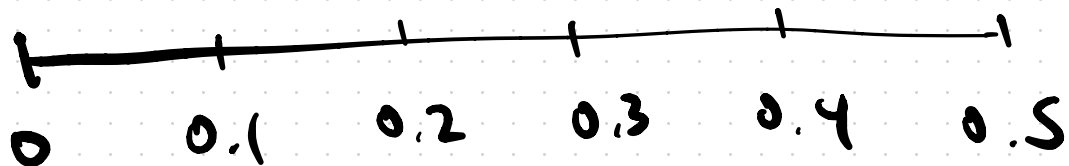
left *right*

About how far has it traveled after half a second?

If something is moving at constant speed,

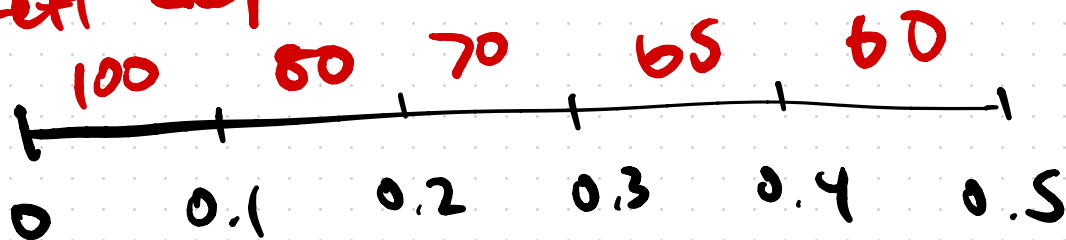
$$\text{distance} = \text{speed} \cdot \text{time}$$

Strategy: Break up 0.5 s into 0.1 s intervals, and assume constant speed on each interval.



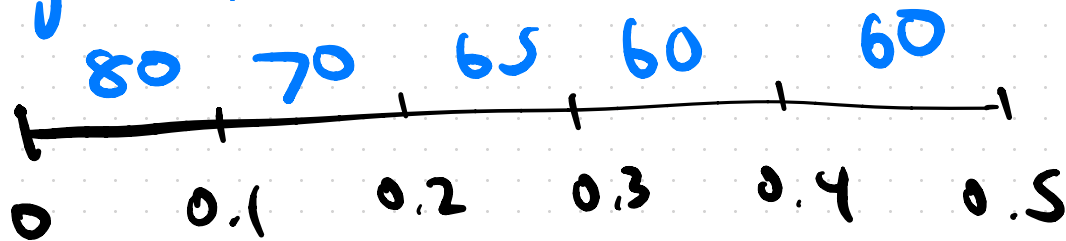
What should we choose for the speed on each interval?

Left endpoints



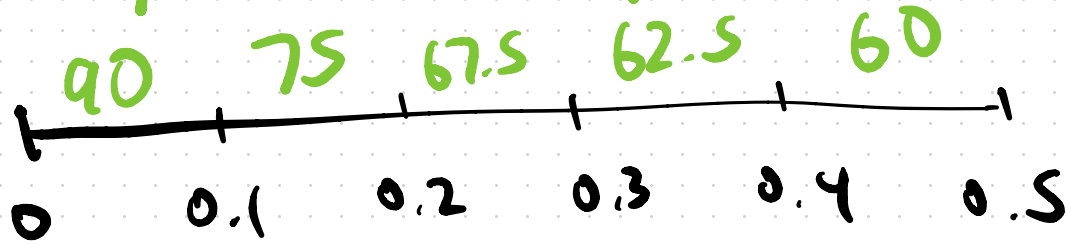
$$\begin{aligned}
 \text{Total distance travelled} &= 100(0.1) + 80(0.1) + 70(0.1) \\
 &\quad + 65(0.1) + 60(0.1) \\
 &= 37.5
 \end{aligned}$$

Right endpoints



$$\begin{aligned}
 \text{Distance} &= 80(0.1) + 70(0.1) + 65(0.1) + 60(0.1) + 60(0.1) \\
 &= 33.5
 \end{aligned}$$

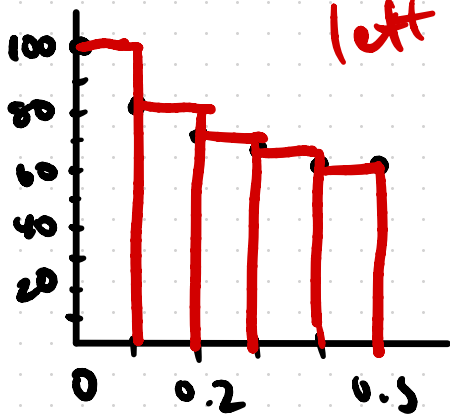
Average of left and right



$$\begin{aligned} \text{Distance} &= 90(0.1) + 75(0.1) + 67.5(0.1) \\ &\quad + 62.5(0.1) + 60(0.1) \\ &= 35.5 \end{aligned}$$

Velocity (m/s)

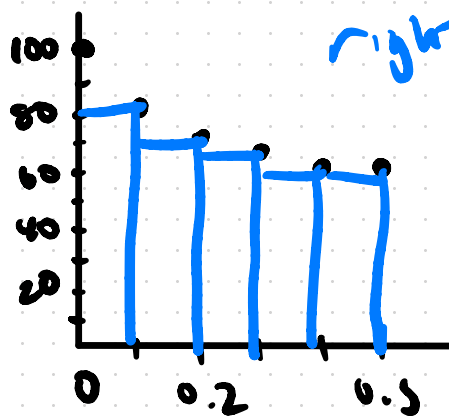
left



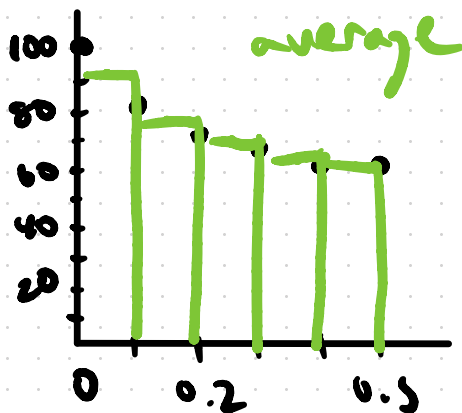
Time(s)

Velocity (m/s)

right

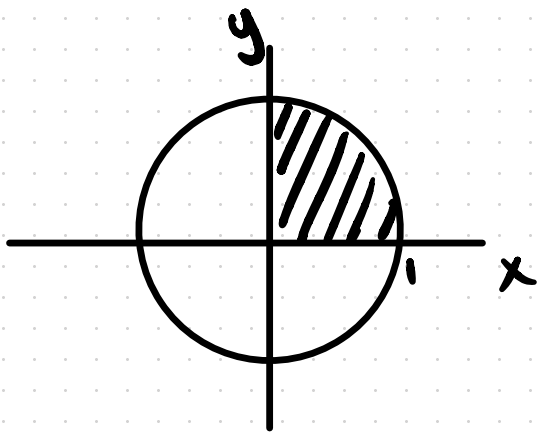


Time(s)

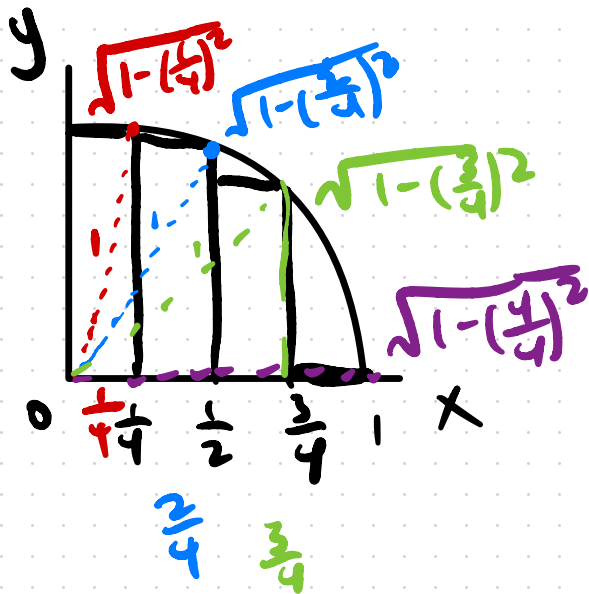


Time(s)

What is the area of a circle with radius 1?



Break up $[0, 1]$
into 4 smaller
intervals



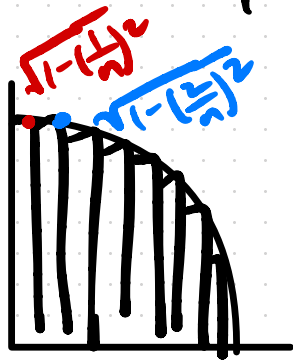
$$y = \sqrt{1-x^2}$$

Area of rectangles =

$$\sqrt{1 - \left(\frac{1}{4}\right)^2} \cdot \frac{1}{4} + \sqrt{1 - \left(\frac{2}{4}\right)^2} \cdot \frac{1}{4} +$$

$$\sqrt{1 - \left(\frac{3}{4}\right)^2} \cdot \frac{1}{4} + \sqrt{1 - \left(\frac{4}{4}\right)^2} \cdot \frac{1}{4}$$

Break up into n smaller intervals, n large



$$\begin{aligned} \text{Area of rectangles} = & \sqrt{1 - \left(\frac{1}{n}\right)^2} \cdot \frac{1}{n} + \sqrt{1 - \left(\frac{2}{n}\right)^2} \cdot \frac{1}{n} \\ & + \dots + \sqrt{1 - \left(\frac{n}{n}\right)^2} \cdot \frac{1}{n} \end{aligned}$$

$$= \sum_{i=1}^n \sqrt{1 - \left(\frac{i}{n}\right)^2} \cdot \frac{1}{n}$$

The area of the sector is the value this sum approaches as $n \rightarrow \infty$, i.e.

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 - \left(\frac{i}{n}\right)^2} \cdot \frac{1}{n}$$